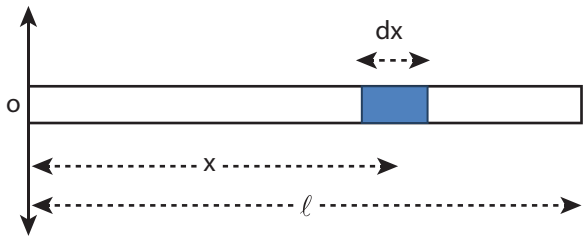
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of this rod, we choose an infinitesimally small mass dm of elemental length dx at a distance x from the origin.

λ is the linear mass density (i.e. mass per unit length) of the rod. $\lambda = \frac{M}{\ell}$

The mass of small element (dm) is, $dm = \frac{M}{\ell} dx$

Now, we can write the center of mass equation for this mass distribution as,

$$x_{CM} = \frac{\int x dm}{\int dm}$$

$$x_{CM} = \frac{\int_0^{\ell} x\left(\frac{M}{\ell} dx\right)}{M} = \frac{1}{\ell}\int_0^{\ell} x dx$$

$$= \frac{1}{\ell}\left[\frac{x^2}{2}\right]_0^{l} = \frac{1}{\ell}\left(\frac{\ell^2}{2}\right)$$

$$x_{CM} = \frac{\ell}{2}$$

As the position $\frac{\ell}{2}$ is the geometric center of the rod, it is concluded that the center of mass of the uniform rod is located at its geometric center itself.

5.1.6 Motion of Center of Mass

When a rigid body moves, its center of mass will also move along with the body. For kinematic quantities like velocity (v_{CM}) and acceleration (a_{CM}) of the center of mass, we can differentiate the expression for position of center of mass with respect to time once and twice respectively. For simplicity, let us take the motion along X direction only.

$$\vec{v}_{CM} = \frac{d\vec{x}_{CM}}{dt} = \frac{\sum m_i\left(\frac{d\vec{x}_i}{dt}\right)}{\sum m_i}$$

$$\vec{v}_{CM} = \frac{\sum m_i \vec{v}_i}{\sum m_i} \qquad (5.7)$$

$$\vec{a}_{CM} = \frac{d}{dt}\left(\frac{d\vec{x}_{CM}}{dt}\right) = \left(\frac{d\vec{v}_{CM}}{dt}\right) = \frac{\sum m_i\left(\frac{d\vec{v}_i}{dt}\right)}{\sum m_i}$$

$$\vec{a}_{CM} = \frac{\sum m_i \vec{a}_i}{\sum m_i} \qquad (5.8)$$

In the absence of external force, i.e. $\vec{F}_{ext} = 0$, the individual rigid bodies of a system can move or shift only due to the internal forces. This will not affect the position of the center of mass. This means that the center of mass will be in a state of rest or uniform motion. Hence, $\vec{v}_{CM}$ will be zero when center of mass is at rest and constant when center of mass has uniform motion $(\vec{v}_{CM} = 0$ or $\vec{v}_{CM} = \text{constant})$. There will be no acceleration of center of mass, $(\vec{a}_{CM} = 0)$.